



VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

CASE STUDY -1

NAME: POKURI YASWANTH CHOWDARY

REG.NO: 23MID0051

COURSE CODE: CSI3005

COURSE TITLE: ADVANCED DATA VISUALIZATION TECHNIQUES

FACULTY: RAMESH C

SLOT: A1+TA1

CLASS NUMBER: VL2025260501536

PROGRAMME: M. TECH (INTEGRATED) - CSE

SEMESTER: WINTER 2025-26

SCHOOL: SCOPE

EMAIL ID: pokuri.yaswanth2023@vitstudent.ac.in

GitHub Repo Link: <https://github.com/chowdary0708/chess-data-visualization-dashboard.git>

[Q1] Dashboard Creation for a Given Dataset:

You have been assigned a unique dataset (available in Moodle). Your task is to design and develop an interactive dashboard using Tableau / Power BI / Shiny to effectively analyze the data and extract relevant business insights for decision-making.

Note: The completed work may also be uploaded to **GitHub**, and the repository link should be included in the submitted files.

Tasks:

1. **Data Exploration & Preparation**
 - Examine your dataset and identify key dimensions and measures.
 - Perform necessary preprocessing (handle missing values, outliers, and formatting).
2. **Dashboard Development**
 - Design and create a dashboard that includes:
 - Time-series analysis (Sales/Profit trends over time)
 - Category-wise and region-wise comparisons
 - Distribution of key variables
 - Relationship between variables (e.g., Sales vs Profit)
3. **Visualization Justification**
 - Select appropriate charts and provide justification for each visualization used.
4. **Interactivity**
 - Implement interactive features such as:
 - Filters (Date, Category, Region)
 - Drill-down capabilities
 - Tooltips
5. **Insight Generation**
 - Identify and report:
 - Key trends and patterns
 - Best and worst performing categories/regions
 - Any anomalies or unusual observations
6. **Performance Optimization**
 - Apply suitable techniques to handle large datasets efficiently.

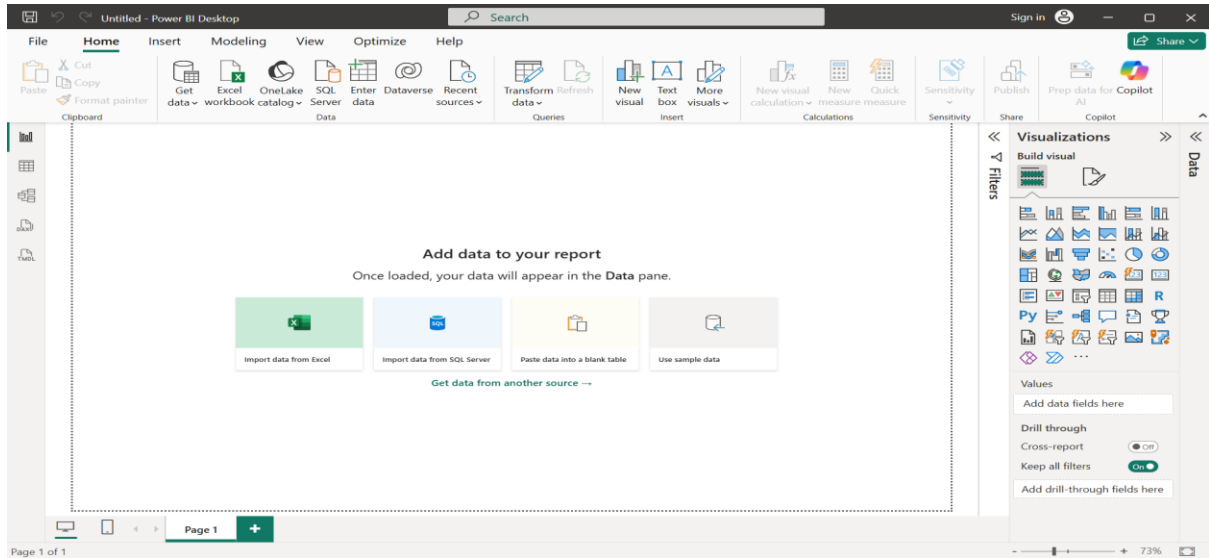
Dataset:`chess_games.csv`

A) DATA EXPLORATION & PREPARATION

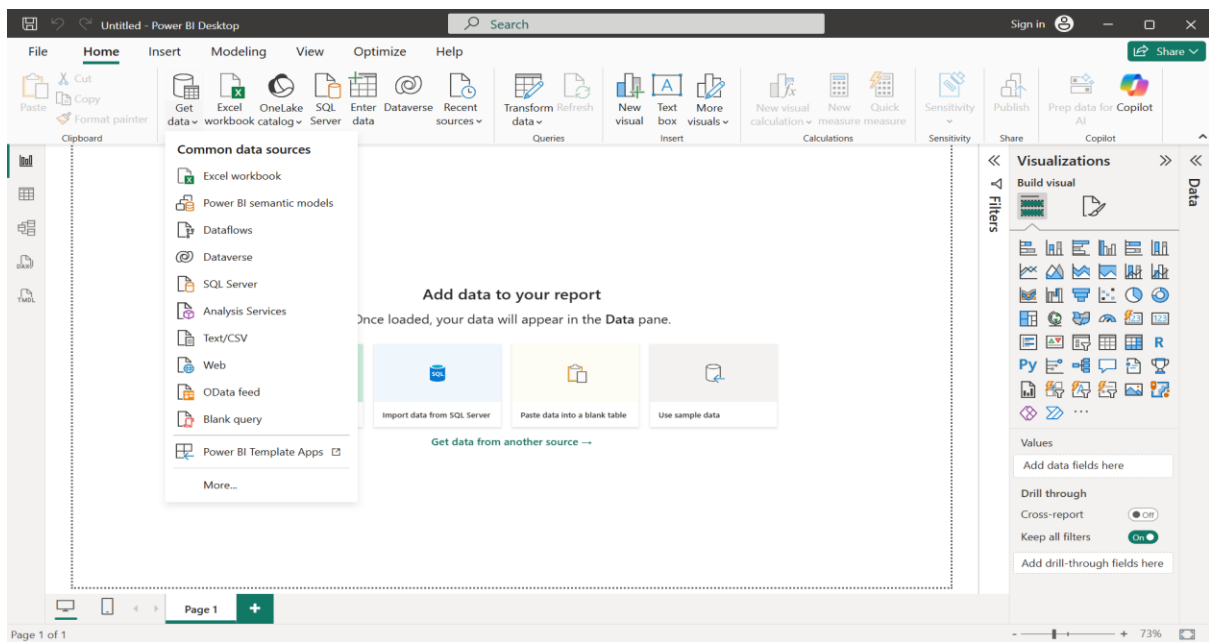
1.Data Loading

The dataset was imported into Microsoft Power BI using the Get Data option. The Transform Data option was selected to clean and prepare the data before analysis.

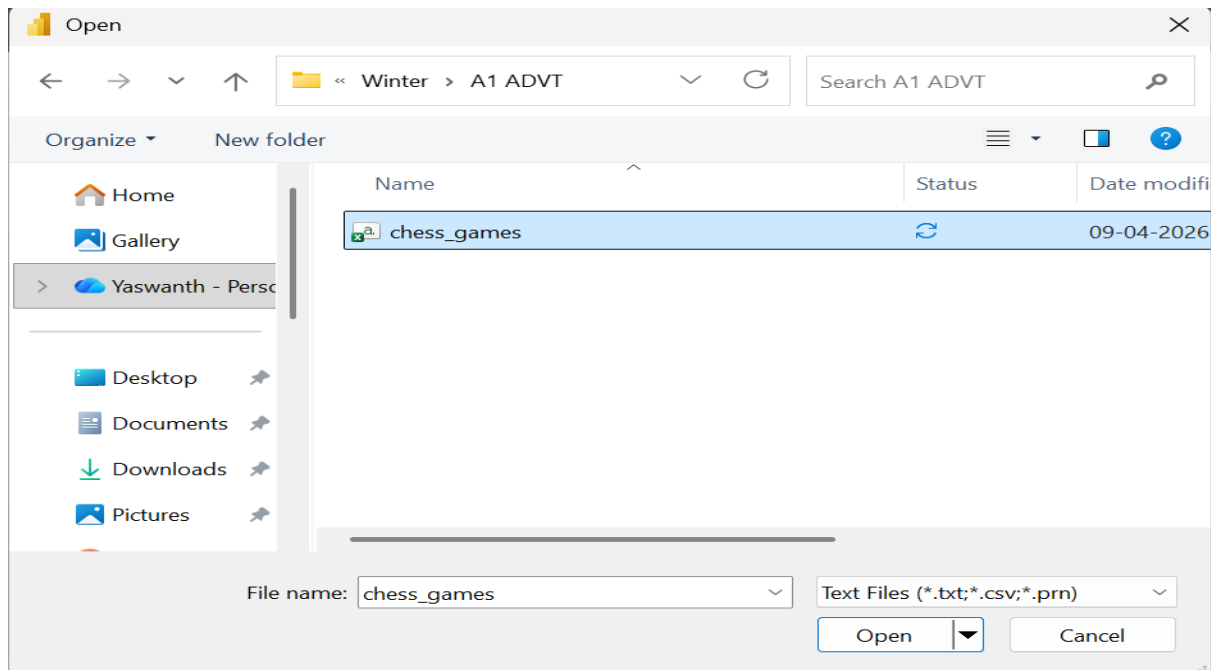
Open “PowerBI Desktop”:



Click on “Get Data” and select respective file type of dataset:



Select the respective dataset:



Click on “Transform”:

chess_games.csv

File Origin

1252: Western European (Windows)

Delimiter

Comma

Data Type Detection

Based on first 200 rows

game_id	rated	turns	victory_status	winner	time_increment	white_id	white_rating	black_id	black_rating
1	FALSE	13	Out of Time	White	15+2	bourgris	1500	a-00	1191
2	TRUE	16	Resign	Black	5+10	a-00	1322	skinnerua	1261
3	TRUE	61	Mate	White	5+10	ischia	1496	a-00	1500
4	TRUE	61	Mate	White	20+0	daniamurashov	1439	adivanov2009	1454
5	TRUE	95	Mate	White	30+3	nik221107	1523	adivanov2009	1469
6	FALSE	5	Draw	Draw	10+0	trelynn17	1250	franklin14532	1002
7	TRUE	33	Resign	White	10+0	capa_jr	1520	daniel_likes_chess	1423
8	FALSE	9	Resign	Black	15+30	daniel_likes_chess	1413	soultego	2108
9	TRUE	66	Resign	Black	15+0	ehabfanri	1439	daniel_likes_chess	1392
10	TRUE	119	Mate	White	10+0	daniel_likes_chess	1381	mirco25	1209
11	FALSE	39	Mate	White	20+60	daniel_likes_chess	1381	anaissac	1272
12	FALSE	38	Resign	Black	20+60	daniel_likes_chess	1381	subham777	1867
13	FALSE	60	Resign	Black	5+40	daniel_likes_chess	1381	roman12342005	1936
14	FALSE	31	Resign	Black	8+0	daniel_likes_chess	1381	alkhan	1607
15	FALSE	31	Mate	White	15+15	shivangithegenius	1094	sureka_akshat	1141
16	FALSE	43	Resign	Black	15+15	sureka_akshat	1141	shivangithegenius	1094
17	FALSE	52	Resign	Black	15+15	shivangithegenius	1094	sureka_akshat	1141
18	FALSE	66	Mate	Black	15+16	sureka_akshat	1141	shivangithegenius	1094
19	FALSE	101	Resign	Black	15+15	shivangithegenius	1094	slam_ment	1300
20	FALSE	25	Resign	White	11+0	g-ios	1500	shivangithegenius	1094

Extract Table Using Examples

Load

Transform Data

Cancel

2. Understanding the Data

The dataset contained country details and multiple indicators like HDI and Life Expectancy across different years. The data was in wide format, making it difficult to analyze.

Table: TransformColumnTypes(#"Promoted Headers",{"game_id", Int64.Type}, {"rated", type logical})

game_id	rated	turns	victory_status	winner
1	FALSE	13	Out of Time	White
2	TRUE	16	Resign	Black
3	TRUE	61	Mate	White
4	TRUE	61	Mate	White
5	TRUE	95	Mate	White
6	FALSE	5	Draw	Draw
7	TRUE	33	Resign	White
8	FALSE	9	Resign	Black
9	TRUE	66	Resign	Black
10	TRUE	119	Mate	White
11	FALSE	39	Mate	White
12	FALSE	38	Resign	Black
13	FALSE	60	Resign	Black
14	FALSE	31	Resign	Black
15	FALSE	31	Mate	White
16	FALSE	43	Resign	Black
17	FALSE	52	Resign	Black
18	FALSE	66	Mate	Black
19	FALSE	101	Resign	Black
20	FALSE	25	Resign	White
21	FALSE	14	Resign	Black
22	FALSE	3	Resign	White
23	FALSE	17	Resign	White
24	TRUE	36	Resign	White

3. Unpivoting the Data

Selected the columns white_rating and black_rating and applied Unpivot Columns to convert them into two columns: Attribute and Value.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Data Type: Text Replace Values Unpivot Columns

Detect Data Type Fill Move

Pivot Column Convert to List

Split Column Format Text Column

Merge Columns

Statistics Standard Scientific

Trigonometry

10²

Rounding

Information

Date Time Duration

Run R script Run Python script

Queries [1]

chess_games

Table

opening_shortname opening_response opening_variation Attribute Value

1	Defense		Exchange Variation	white_rating	1500
2	Defense		Exchange Variation	black_rating	1191
3	izowitsch Defense		Kennedy Variation	white_rating	1322
4	izowitsch Defense		Kennedy Variation	black_rating	1261
5	g's Pawn Game		Leonardis Variation	white_rating	1496
6	g's Pawn Game		Leonardis Variation	black_rating	1500
7	en's Pawn Game		Zukertort Variation	white_rating	1439
8	en's Pawn Game		Zukertort Variation	black_rating	1454
9	idor Defense			white_rating	1523
10	idor Defense			black_rating	1469
11	ian Defense		Mongoose Variation	white_rating	1250
12	ian Defense		Mongoose Variation	black_rating	1002
13	ikmar-Diemer Gambit		Pietrowsky Defense	white_rating	1520
14	ikmar-Diemer Gambit		Pietrowsky Defense	black_rating	1423
15	izowitsch Defense		Kennedy Variation	white_rating	1413
16	izowitsch Defense		Kennedy Variation	black_rating	2108
17	an Game		Schilling-Kostic Gambit	white_rating	1439
18	an Game		Schilling-Kostic Gambit	black_rating	1392
19	ndinavian Defense		Mieses-Kotroc Variation	white_rating	1381
20	ndinavian Defense		Mieses-Kotroc Variation	black_rating	1209
21	t Kruijs Opening			white_rating	1381
22	t Kruijs Opening			black_rating	1272
23	ch Defense		Advance Variation	white_rating	1381
24	ch Defense		Advance Variation	black_rating	1867
25					

17 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 21:50

Query Settings

PROPERTIES

Name chess_games

APPLIED STEPS

Source

Promoted Headers

Unpivoted Columns

Changed Type

4. Renaming Columns

Renamed the Attribute column to Player_Type and Value column to Rating for better understanding and readability of the dataset.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Data Type: Text Replace Values Unpivot Columns

Detect Data Type Fill Move

Pivot Column Convert to List

Split Column Format Text Column

Merge Columns

Statistics Standard Scientific

Trigonometry

10²

Rounding

Information

Date Time Duration

Run R script Run Python script

Queries [1]

chess_games

Table

opening_shortname opening_response opening_variation Player_Type Rating

1	Defense		Exchange Variation	white_rating	1500
2	Defense		Exchange Variation	black_rating	1191
3	izowitsch Defense		Kennedy Variation	white_rating	1322
4	izowitsch Defense		Kennedy Variation	black_rating	1261
5	g's Pawn Game		Leonardis Variation	white_rating	1496
6	g's Pawn Game		Leonardis Variation	black_rating	1500
7	en's Pawn Game		Zukertort Variation	white_rating	1439
8	en's Pawn Game		Zukertort Variation	black_rating	1454
9	idor Defense			white_rating	1523
10	idor Defense			black_rating	1469
11	ian Defense		Mongoose Variation	white_rating	1250
12	ian Defense		Mongoose Variation	black_rating	1002
13	ikmar-Diemer Gambit		Pietrowsky Defense	white_rating	1520
14	ikmar-Diemer Gambit		Pietrowsky Defense	black_rating	1423
15	izowitsch Defense		Kennedy Variation	white_rating	1413
16	izowitsch Defense		Kennedy Variation	black_rating	2108
17	an Game		Schilling-Kostic Gambit	white_rating	1439
18	an Game		Schilling-Kostic Gambit	black_rating	1392
19	ndinavian Defense		Mieses-Kotroc Variation	white_rating	1381
20	ndinavian Defense		Mieses-Kotroc Variation	black_rating	1209
21	t Kruijs Opening			white_rating	1381
22	t Kruijs Opening			black_rating	1272
23	ch Defense		Advance Variation	white_rating	1381
24	ch Defense		Advance Variation	black_rating	1867
25					

17 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 21:50

Query Settings

PROPERTIES

Name chess_games

APPLIED STEPS

Source

Promoted Headers

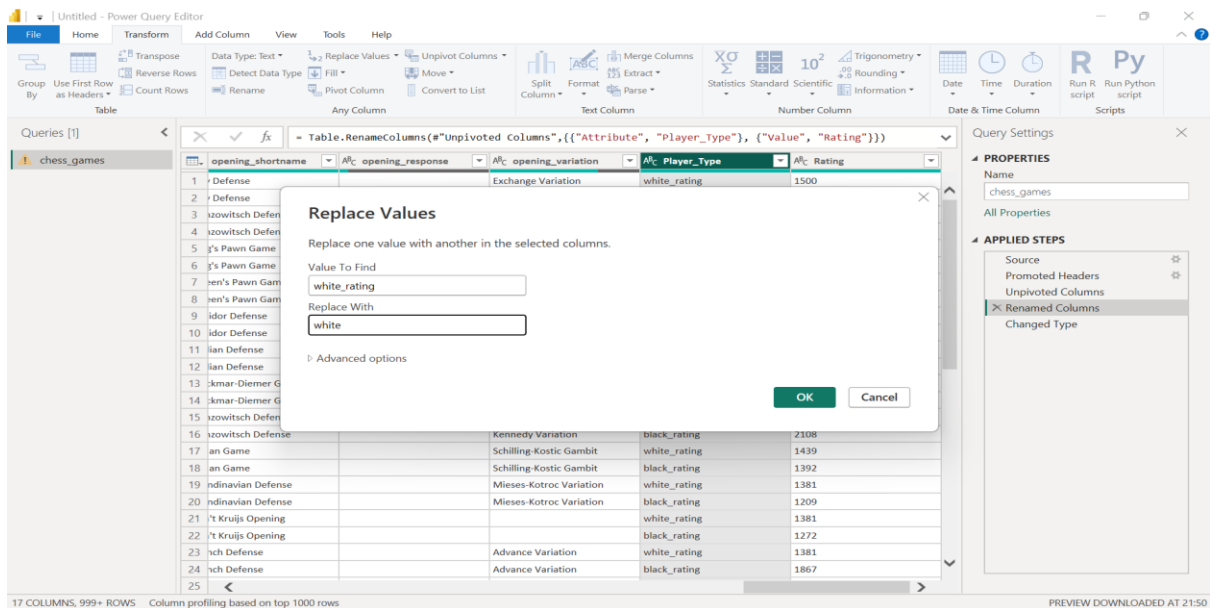
Unpivoted Columns

Renamed Columns

Changed Type

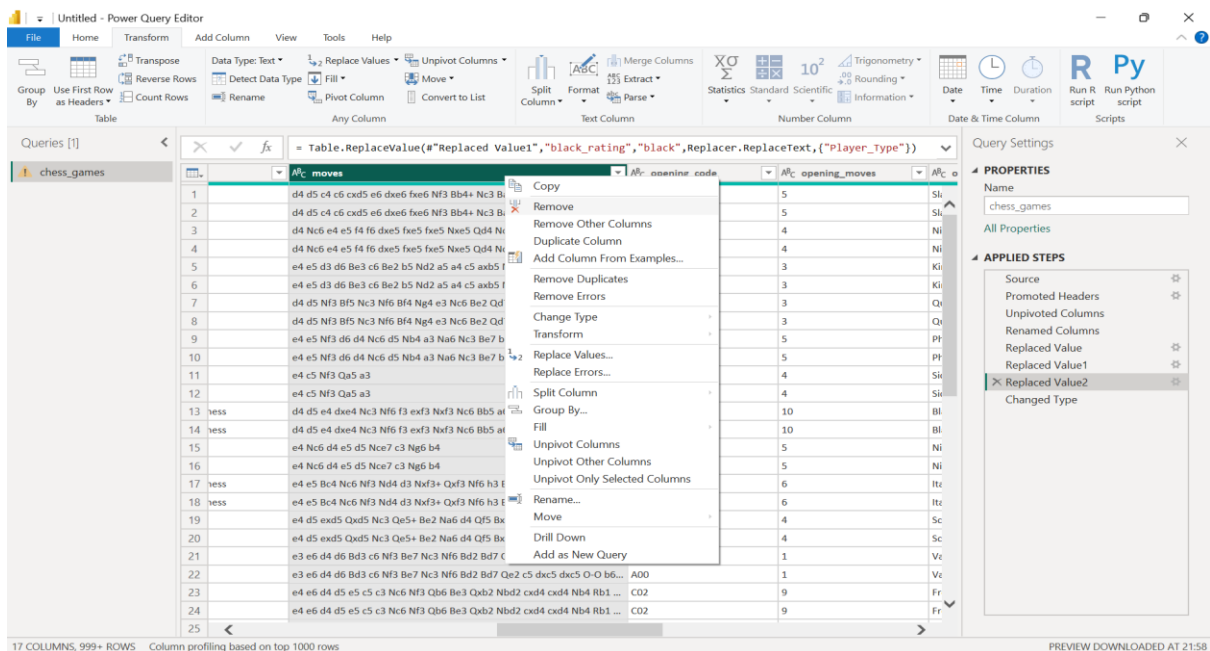
5. Cleaning The Player_Type Column

Cleaned the Player_Type column by replacing values “white_rating” and “black_rating” with “white” and “black” respectively to improve readability and simplify analysis.



6. Remove Unnecessary Columns

Removed unnecessary columns such as moves and other irrelevant fields to reduce dataset size and improve performance. Only required columns like game_id, Player_Type, Rating, winner, and opening_name were retained.



7. Fix Data Types

Converted Rating column to decimal format and ensured all other columns had appropriate data types such as text and whole number for accurate analysis.

Microsoft Power Query Editor interface showing a table of chess games. The table has columns: opening_fullname, Player_Type, and Rating. The 'Data Type' dropdown menu is open, showing various data types. The 'Removed Columns' step is highlighted in the 'Applied Steps' pane on the right.

opening_fullname	Player_Type	Rating
Slav Defense: Exchange Variation	white	1500
Slav Defense: Exchange Variation	black	1191
Nimzowitsch Defense: Kennedy Variation	white	1322
Nimzowitsch Defense: Kennedy Variation	black	1261
King's Pawn Game: Leonardi Variation	white	1496
King's Pawn Game: Leonardi Variation	black	1500
Queen's Pawn Game: Zukertort Variation	white	1439
Queen's Pawn Game: Zukertort Variation	black	1454
Philidor Defense	white	1523
Philidor Defense	black	1469
Sicilian Defense: Mongoose Variation	white	1250
Sicilian Defense: Mongoose Variation	black	1002
Blackmar-Diemer Gambit: Pietrowsky Defense	white	1520
Blackmar-Diemer Gambit: Pietrowsky Defense	black	1423
Nimzowitsch Defense: Kennedy Variation Linksspringer Variation	white	1413
Nimzowitsch Defense: Kennedy Variation Linksspringer Variation	black	2108
Italian Game: Schilling-Kostic Gambit	white	1439
Italian Game: Schilling-Kostic Gambit	black	1392
Scandinavian Defense: Mieses-Kotroc Variation	white	1381
Scandinavian Defense: Mieses-Kotroc Variation	black	1209
Van't Kruijs Opening	white	1381
Van't Kruijs Opening	black	1272
French Defense: Advance Variation Paulsen Attack	white	1381
French Defense: Advance Variation Paulsen Attack	black	1867

B) DASHBOARD DEVELOPMENT

An interactive dashboard was created in **Microsoft Power BI** to analyze the chess dataset and extract meaningful insights.

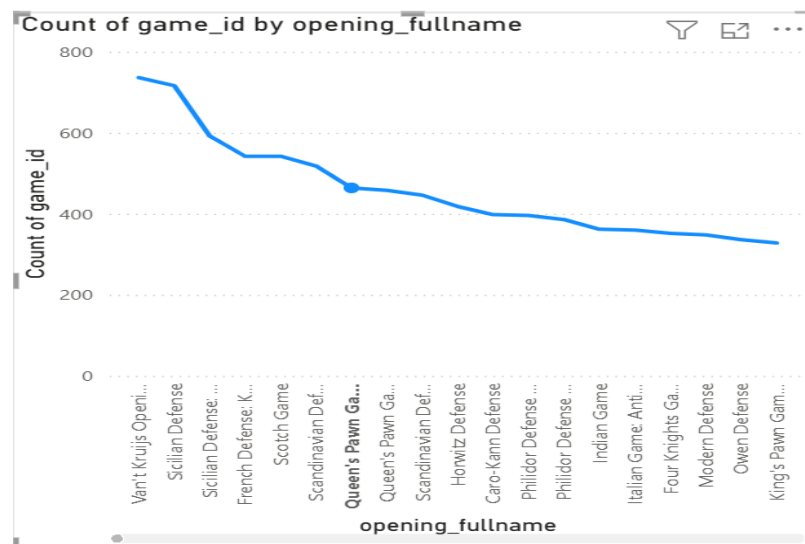
1.Time-Series Analysis (Line Chart)

To show how the number of chess games varies over time.

i) Click **Line Chart** icon in the Visualizations panel

ii) Drag fields:

- game_id → Y-axis (set to Count)
- opening_fullname → X-axis



2) Category-wise & Region-wise Comparison (Bar Chart)

Compare average ratings across different openings

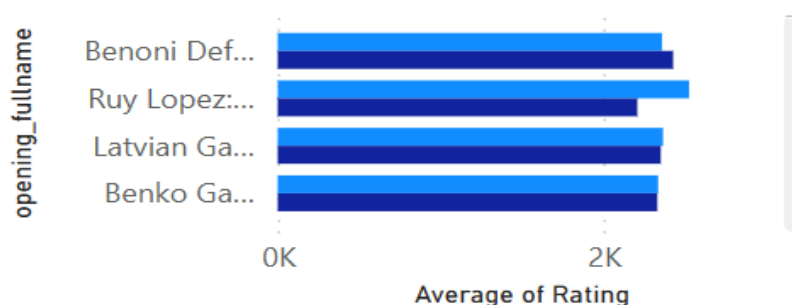
1. Click Clustered Bar Chart (

2. Drag:

- opening_fullname → Axis
- Rating → Values
- Player_Type → Legend

Average of Rating by opening_fullname and Player_Type

Player_Type ● black ● white



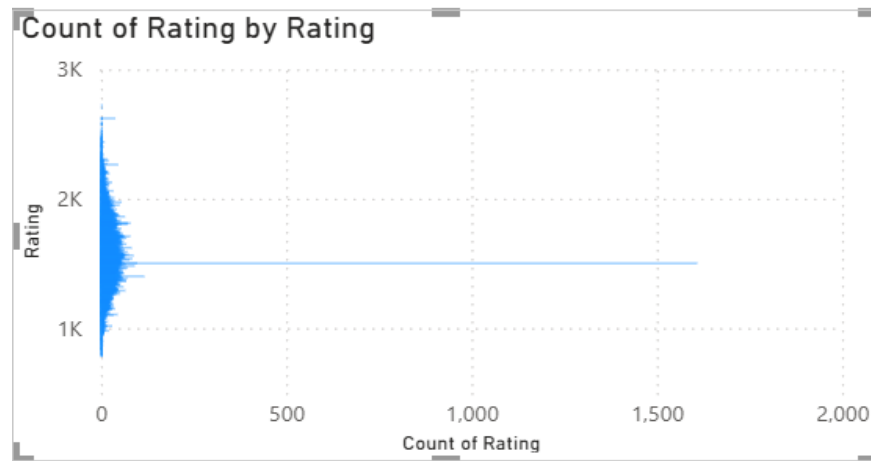
3) Distribution of Variables (Column Chart)

Understand how ratings are distributed (low, medium, high)

1. Click Clustered Column Chart

2. Drag:

- Rating → Axis
- Rating → Values



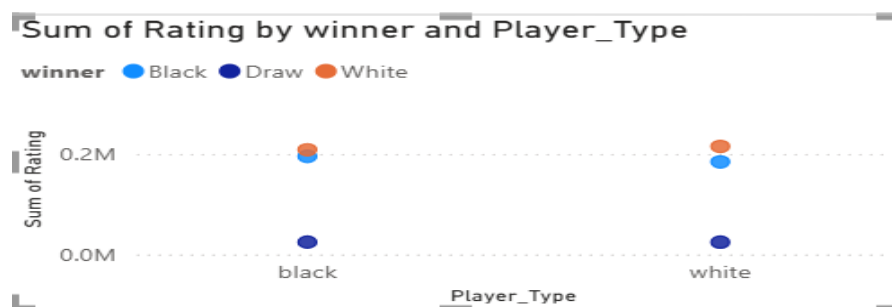
4) Relationship Between Variables (Scatter Plot)

Show relationship between: Player_Type vs Rating

1. Click Scatter Chart

2. Drag:

- Player_Type → X-axis
- Rating → Y-axis
- winner → Legend



C) VISUALIZATION JUSTIFICATION

1. Line Chart (Trend Analysis)

A line chart was selected to analyze the frequency of chess openings because it effectively represents patterns across different categories. By placing opening names on the X-axis and the count of games on the Y-axis, the chart clearly shows which openings are most commonly played. The continuous structure of the line helps in identifying variations and trends across different openings, making it useful for comparative analysis.

2. Bar Chart (Category-wise Comparison)

A bar chart was used to compare average player ratings across different chess openings because it provides a clear comparison between categories. By placing openings on the axis and average rating on the value field, the chart highlights which openings are associated with higher or lower ratings. The addition of Player_Type as a legend enhances the comparison by distinguishing between white and black players.

3. Column Chart (Distribution of Variables)

A column chart was used to analyze the distribution of player ratings. This visualization helps in understanding how ratings are spread across different ranges. By using rating as both the axis and value (count), the chart shows the frequency of ratings, making it easy to identify common rating levels and patterns in the dataset.

4. Scatter Plot (Relationship Between Variables)

A scatter plot was chosen to examine the relationship between player type and rating. By plotting player type on one axis and rating on the other, the chart helps in identifying variations and patterns in player performance. It is useful for detecting clusters and differences between player categories.

D) INTERACTIVITY

Interactive features were implemented in the dashboard to allow users to explore the data dynamically and gain deeper insights. These features improve usability by enabling filtering, detailed analysis, and better data understanding.

1. Filters (Slicers)

The Slicer visual was used in Microsoft Power BI Added slicers for:

- opening_fullname
- Player_Type

- Player_Type

opening_fullname

- ☐ Alekhine Defense
- ☐ Alekhine Defense #2
- ☐ Alekhine Defense #3
- ☐ Alekhine Defense: Balogh Variation
- ☐ Alekhine Defense: Brooklyn Variation
- ☐ Alekhine Defense: Exchange Variation
- ☐ Alekhine Defense: Four Pawns Attack
- ☐ Alekhine Defense: Four Pawns Attack...

Player_Type

- ☐ black
- ☐ white

winner

- ☐ Black
- ☐ Draw
- ☐ White

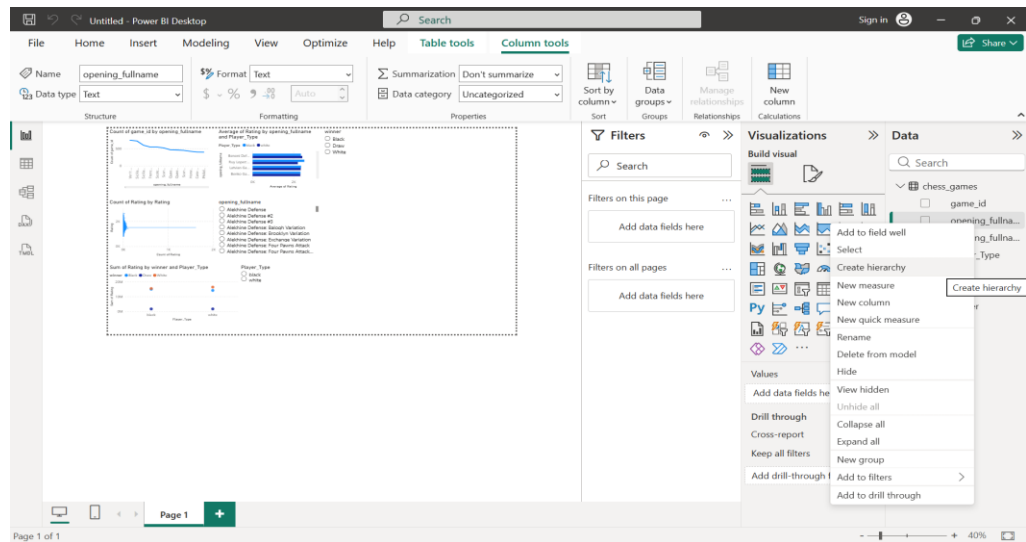
2. Drill-down

Use your Bar Chart (opening_fullname vs Rating)

1. Create Hierarchy

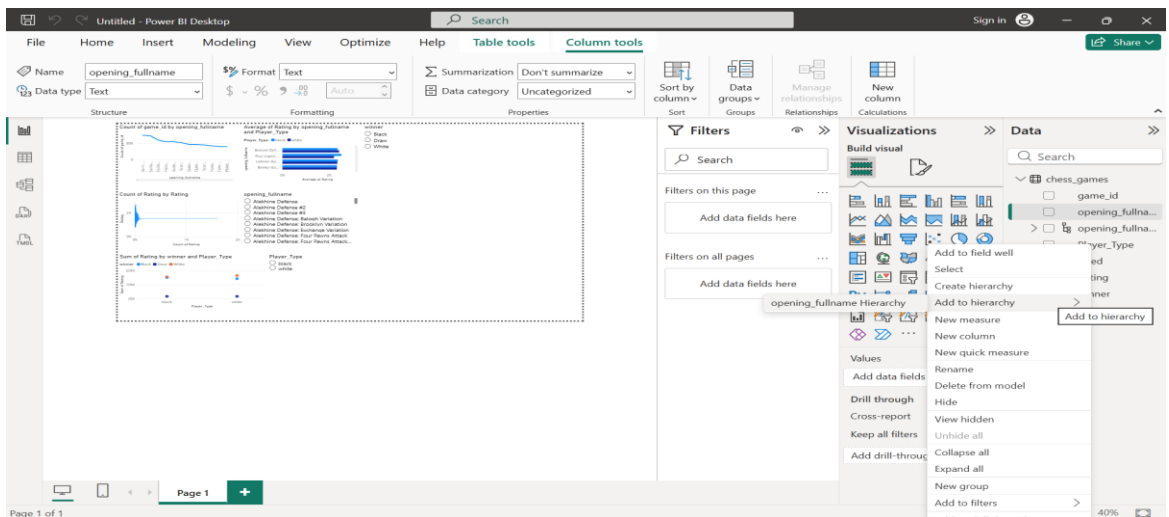
1. In Data panel:

- o Right-click opening_fullname and click Create Hierarchy.



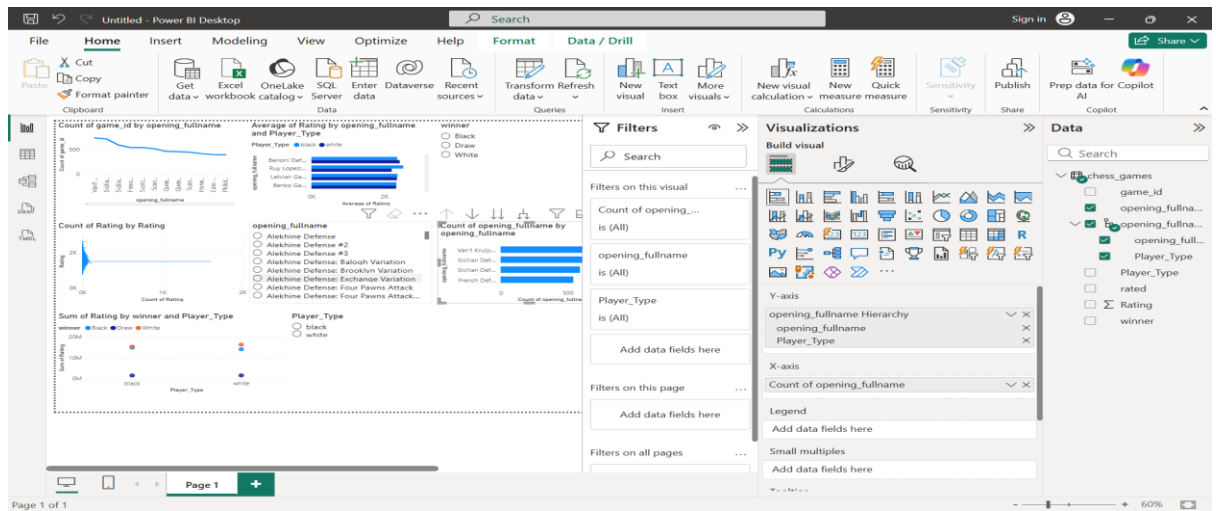
2. Add:

Drag Player_Type into the same hierarchy



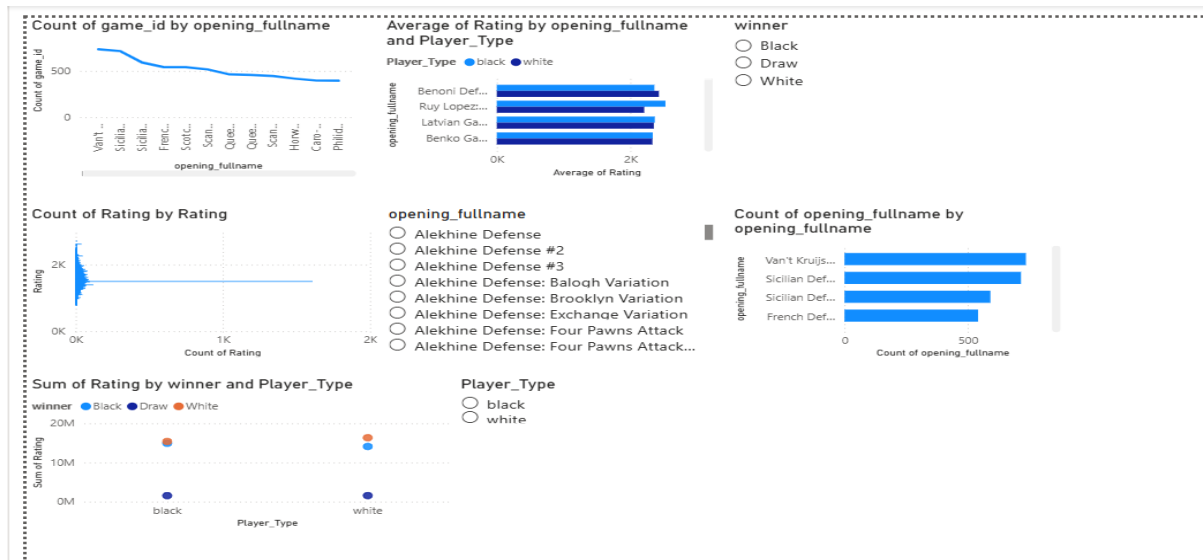
2. Apply to chart

1. Drag this hierarchy into Axis

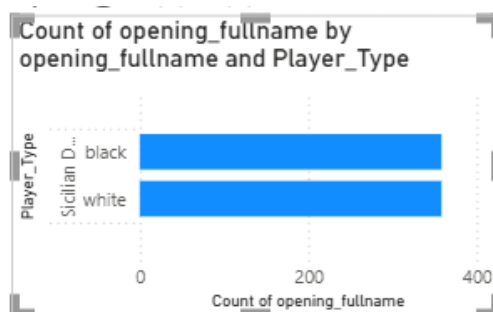


2. Click Drill-down icon on chart

Before clicking “Drill down”:



After clicking “Drill down”:



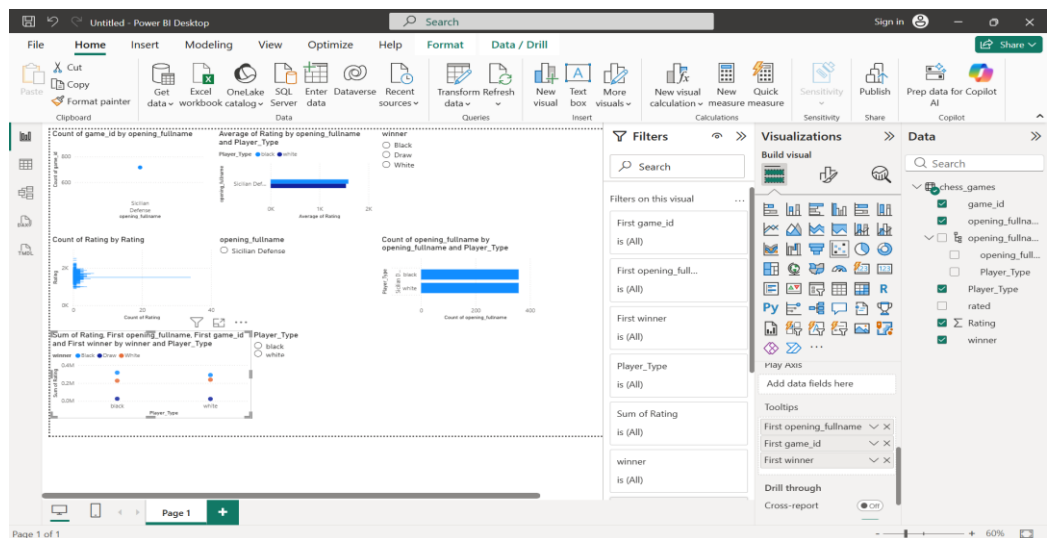
3.ToolTip

Use your Scatter Plot (Rating vs Player_Type)

1. Add Fields to Tooltips

In Visualizations panel:

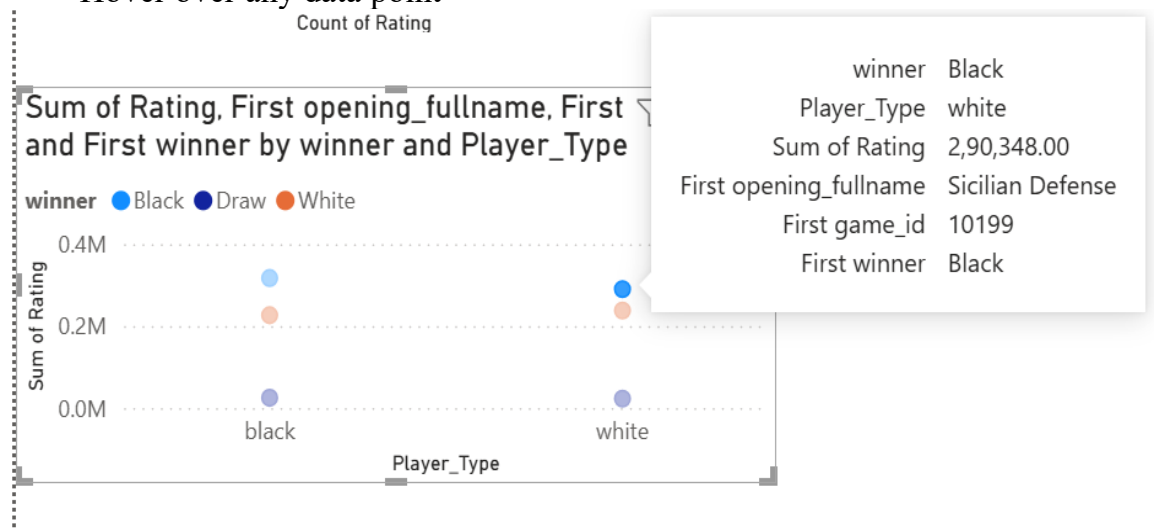
- Click the Scatter Plot Add:
- Drag opening_fullname → Tooltips
- Drag winner → Tooltips
- (Optional) Drag game_id → Tooltips



2. Check Tooltip

Move cursor on chart:

- Hover over any data point



E) INSIGHT GENERATION

The dashboard was analyzed to identify key trends, performance differences, and unusual patterns in the chess dataset.

1. Key Trends and Patterns

- Certain chess openings are played more frequently, indicating strong player preference for specific strategies.
- The rating distribution shows that most players fall within a mid-range, with fewer players at extreme high or low ratings.
- There is a variation between **white and black player types**, showing differences in gameplay and outcomes.
- Openings associated with higher average ratings are likely preferred by more experienced players.

2. Best and Worst Performing Categories

- **Best Performing:**
Openings with higher average ratings indicate strong performance and are typically used by skilled players. These openings consistently show higher rating values.
- **Moderate Performing:**
Some openings show average ratings, indicating balanced performance and usage among intermediate players.
- **Worst Performing:**
Openings with lower average ratings suggest lower performance or usage by less experienced players.

3. Anomalies and Unusual Observations

- Some data points show unusually high or low ratings, indicating outliers in player performance.
- Certain openings have inconsistent ratings, showing variability in effectiveness.
- A few data points deviate from general patterns, suggesting unexpected gameplay outcomes.

F) PERFORMANCE OPTIMIZATION

Several techniques were applied to improve the performance of the dashboard and ensure efficient handling of the dataset.

Techniques Used:

- **Removed Unnecessary Columns:**
Only essential columns such as game_id, Player_Type, Rating, winner, and opening_fullname were retained to reduce dataset size.
- **Filtered Relevant Data:**
Irrelevant data and unused attributes were removed to focus only on meaningful information.
- **Used Aggregations:**
Aggregations such as **count** and **average** were used to summarize data efficiently and improve performance.
- **Limited Number of Visuals:**
Only necessary charts were included in the dashboard to avoid overloading and improve responsiveness.
- **Optimized Data Types:**
Columns were converted to appropriate data types (e.g., decimal and text) to ensure faster processing and accurate analysis.